Waste and the Waste Stream

Why do we waste things?
- It's no more useful
- It's unwanted
- It's defective
- It's old & worthless

Waste: unwanted by-products and residues in nearly everything we do.

Waste Stream: is the term that describes the steady flow of varied wastes that we all produce, from domestic garbage and yard wastes to industrial, commercial, and construction refuse.

3

Waste and the Waste Stream

Waste streams can be divided into two broad types:

✓ Material-related streams (such as metals or plastics)
✓ Product-related streams (such as electronic waste or end-of-life vehicles)



eurostat

4

Waste and the Waste Stream

Sources of Waste – Global Classification

❑ Household – MSW domestic

❑ Public spaces – MSW public – public bins, legal dumping activities etc.

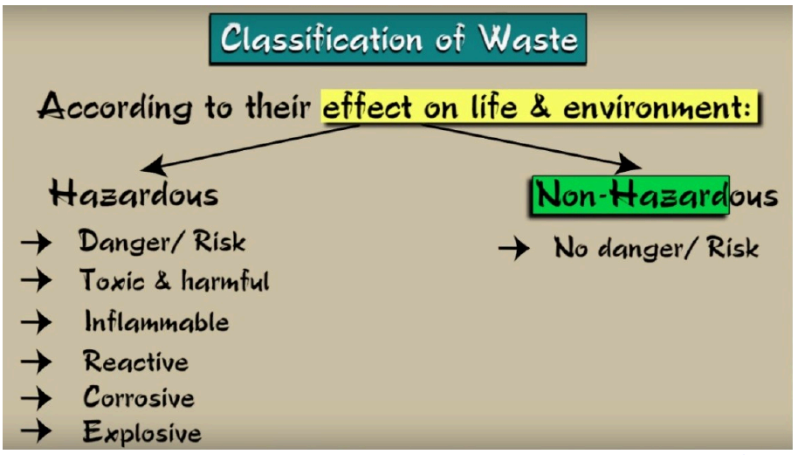
❑ Business – C&I – waste generated by business and industrial productions

❑ Construction – construction and demolition residual

❑ Disaster – waste generated from unplanned events (e.g., flood, cyclone, earthquake, fire hazard etc.) Mandalay

5

Waste and the Waste Stream



Letstute

8

Waste and the Waste Stream



Letstute

9

Waste and the Waste Stream



Letstute

10

Waste Management

The **'polluter pays principle'** requires polluters to bear the cost of preventing, controlling and cleaning up pollution, to ensure that these costs are reflected in the price of goods and services causing pollution at the production and/or consumption stage.

Implementing these principles, **extended producer responsibility (EPR)** is used in several waste Streams. It implies that producers take over the (financial and/or
operational) responsibility for collecting or taking back used goods, for sorting them and for their eventual recycling.

Three 'R's:

Modern waste management method follows the three 'R's which means

- **Reduce -** the amount and toxicity of trash that discard.
- **Reuse -** containers and products; repairing what is broken or giving it to someone who can repair it.
- **Recycle -** as much as possible, which means buying products with recycled content.

15

Waste Management

think about the **benefits** and challenges of **3R**

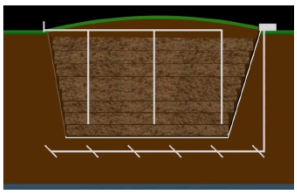


16

Waste Management

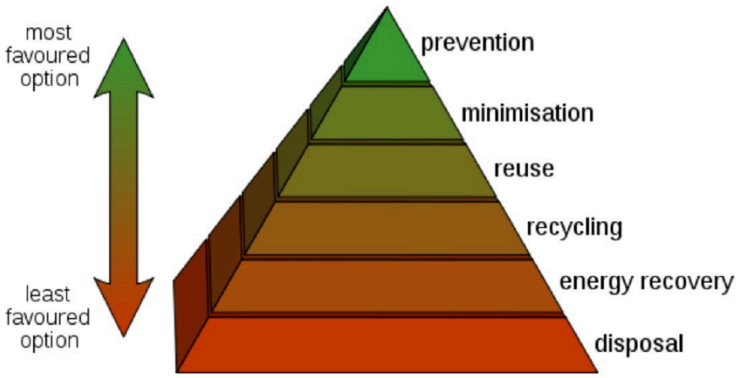
Disposal Methods:

Open dump – drop it some place **Not desirable**
Land fill – dig, fill, separate, compact, secure & cover **Most used**
Ocean dump – dump in ocean **Not desirable**
Incineration – burn & recover **Used**


17

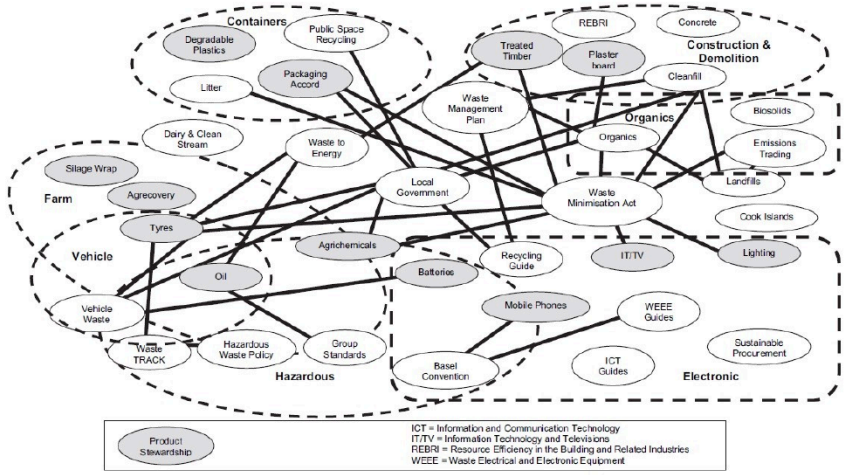
Sustainable Waste Management



MONICA PIANOSI

22

SWM - Complexity of Waste Management



J.K. Seadon, 2010

23

Sustainable Waste Management

Approaches to look at the issue…**many**

Sustainable Approach by GIZ

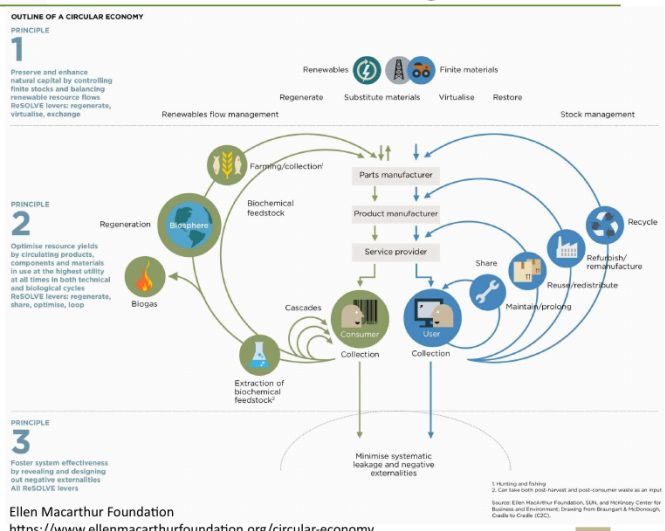
- Waste-to-energy technologies
- Avoidance of marine litter
- Electronic waste
- Economic instruments

giz Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

24

Sustainable Waste Management

Circular Economy Model



Ellen Macarthur Foundation
https://www.ellenmacarthurfoundation.org/circular-economy

28

